INCH–POUND
MIL-DTL-3060G (AR)
w/AMENDMENT 1
10 August 2012
SUPERSEDING
MIL-DTL-3060G (AR)
31 May 2011

DETAIL SPECIFICATION

BOXES, SMALL ARMS AMMUNITION – M19A1, M2A1, AND M2A2

This specification is approved for use by the US Army Armament Research, Development and Engineering Center (ARDEC), and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers three model steel boxes for shipping and storing ammunition (see 6.1).

1.2 Classification. The Boxes are described as follows:

1.2.1 M19A1. The M19A1 is a .18 cubic foot air tight metal can used to store and field ammunition.

1.2.2 M2A1. The M2A1 is a .32 cubic foot air tight metal can used to store and field ammunition.

1.2.3 M2A2. The M2A2 is a .32 cubic foot air tight metal can used to store and field ammunition.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 or 4 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, or 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to the Commander, U.S. Army ARDEC, ATTN: RDAR-QES-E, Picatinny Arsenal, NJ 07806-5000 or emailed to ardecstdzn@conus.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

TT-C-490	-	Chemical Conversion Coatings and Pretreatments for Ferrous Surfaces (Base for Organic Coatings.)

DEPARTMENT OF DEFENSE SPECIFICATION

MIL-A-70625	-	Automated Acceptance Inspection Equipment Design, Testing and Approval, of

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1916	-	DOD Preferred Method for Acceptance of Products
MIL-STD-406	-	Visual Inspection Standards for Terne Plate Cans and Steel Boxes Used in Small Arms Ammunition Packaging
MIL-STD-1168	-	Ammunition Lot Numbering and Ammunition Data Card

(Copies of these documents are available online at https://assist.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Bldg. 4D, Philadelphia, PA 19111-5094)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC) DRAWINGS

7553296	-	Box, Ammunition, M2A1 Assembly
7553315	-	Box, Ammunition, M19A1 Assembly
7553352	-	Box, Shipping; Boxes, Ammunition, M2A1/M2A2
7692103	-	Box, Shipping; Boxes, Ammunition, M19A1
12998172	-	Coatings, High Performance (For Ammunition Containers)
13032542	-	Box, Ammunition, M2A2 Assembly

(Copies of these drawings may be requested through email to pica.drawing.request@conus.army.mil or from U.S. Army ARDEC, ATTN: RDAR-EIS-PE, Picatinny, NJ 07806-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR TESTING AND MATERIALS INTERNATIONAL (ASTM)

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ASTM B 117	-	Standard Practice for Operating Salt Spray (Fog) Apparatus
ASTM D1654	-	Standard Test Method for Evaluation of Painted or Coated Specimens subjected to Corrosive Environments

(Copies of ASTM standards maybe ordered online at http://www.astm.org or from ASTM international, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959).

2.4 Order of precedence. Unless otherwise specified in this specification or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Required inspections.

3.1.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.1.2 Conformance. A sample shall be subjected to conformance inspection in accordance with 4.3.

3.2 Assembly.

3.2.1 M19A1. Components and assemblies for box, small arms ammunition, M19A1 shall be combined to produce an assembly which conforms to the requirements of drawing 7553315.

3.2.2 M2A1. Components and assemblies for Box, Small Arms Ammunition, M2A1 shall be combined to produce an assembly which conforms to the requirements of drawing 7553296.

3.2.3 M2A2. Components and assemblies for box, small arms ammunition, M2A2 shall be combined to produce an assembly which conforms to the requirements of drawing 13032542.

3.3 Protective coating.

3.3.1 Paint. The paint film thickness and coating shall comply with the requirements of drawing 7553296, 7553315, or 13032542 as applicable. In addition, the coating shall be smooth, unbroken and free of blisters, runs, thin spots and foreign matter. Adhesion of the paint to the pretreated surfaces shall comply with the applicable requirements of TT-C-490.

3.3.2 Corrosion resistance. The exterior surfaces of the box shall show no visible evidence of paint blistering, creepage (loss of adhesion), or corrosion of base metal in excess of 1/8 inch on either side of the score marks or rusting of parts of the hardware subject to pressure of abrasion in normal use or exceeding the standards permissible in MIL-STD-406 and ASTM 1654, after exposure to a 5% solution of salt spray for 96 hours. Additionally, for boxes coated in accordance with drawing 12998172, salt spray exposure on 4 X 6 inch coated panels shall be no less than 400 hours.

3.4 Functioning.

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3.4.1 Cover assembly. The cover of the box of the M19A1, M2A1, and M2A2 shall open, close, and be removed without binding or requiring exertion of undue force. The gasket shall not stick to top edges of the box, be cut or split, not shift within the gasket retainer when the cover is opened. Hinge pins shall be secured in body half of hinge. The cover on the M19A1 box assembly only shall not fall off when opened and allowed to hang freely when the box is held in an inverted position. The formed lip on the side of each M19A1 box assembly cover skirt shall catch beneath the edge of the body fold and restrain the cover from opening fully when the partially opened box is inverted.

3.4.2 Hasp and latch. Mating parts of the body hasp and the latch shall meet without requiring deformation of any box part, and the latch shall close and open freely. When closed, the latch of the assembled box shall remain closed until manually opened.

3.4.3 Elevated temperature storage. Following storage of the box in its closed position at an air temperature of not less than 163 degrees Fahrenheit for a period of not less than 24 hours and subsequent return to ambient temperature, the cover of the box shall function as required in 3.4.1 and 3.4.2.

3.5 Airtightness of box assembly. The assembled box shall not release more than four (4) bubbles of air for 15 seconds, or air leakage shall not exceed .4 cubic centimeter per 5 seconds when subjected to an air pressure differential of three (3) pounds per square inch (psi).

3.6 Weld security.

3.6.1 Cover handle assembly. The cover handle of the box shall withstand a pull of 300 pounds for a period of one minute without separation from the cover or failure of either link, clip, or any weld.

3.6.2 Latch, cover and hinge assembly. The latch, cover and hinge assembly shall withstand a pull of 300 pounds (lbs) for the M19A1 box.

3.6.3 Cover hinge assembly. The cover hinge assembly shall withstand a pull of 500 lbs for the M2A1 and M2A2 boxes.

3.6.4 Hasp assembly. The assembly of the hasp to the box shall withstand a pull of 1000 pounds for one minute without failure of the hasp or weld.

3.7 Gasket compression. The boxes shall have a gasket compression 0.090 ± 0.020 inches when closed.

3.7.1 M19A1. The boxes shall have a gasket compression 0.0900±0.020 inches when closed.

3.7.2 M2A1. The boxes shall have a gasket compression 0.0900±0.020 inches when closed.

3.7.3 M2A2. The boxes shall have a gasket compression 0.0900±0.020 inches when closed.

3.8 Workmanship.

3.8.1 Metal defects. The boxes shall be free of burrs, rough spots, sharp projections, cracks, splits, buckles or sintered surfaces.

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3.8.2 Foreign matter. The boxes shall be free of corrosion, chips, dirt, grease and other foreign matter.

4. VERIFICATION

TABLE I. Requirement/verification cross reference.

Method of Verification							Classes of Verification		
N/A – Not Applicable 1 – Analysis 2 – Demonstration 3 – Examination 4 – Test							A – First article B – Conformance		
Section 3 Requirement	Description	Verification Methods					Verification Class		Section 4 Method
		N/A	1	2	3	4	A	B	
3.1.1	First article				X	X	X		4.2, 4.2.1, 4.2.2
3.1.2	Conformance				X	X		X	4.3, 4.3.1, 4.3.2
3.2.1	M19A1				X	X	X	X	4.3.2.1
3.2.2	M2A1				X	X	X	X	4.3.2.1
3.2.3	M2A2				X	X	X	X	4.3.2.1
3.3.1	Paint				X	X	X	X	4.4.1
3.3.2	Corrosion resistance 1/				X	X	X	X	4.4.2, 4.4.2.1
3.4.1	Cover assembly				X		X	X	4.3.2.2, 4.4.3, 4.4.3.b, 4.4.3.c
3.4.2	Hasp and latch				X		X	X	4.3.2.2, 4.4.3, 4.4.3.a
3.4.3	Elevated temperature storage					X	X	X	4.3.2.2, 4.4.3, 4.4.3.d
3.5	Airtightness of box assembly					X	X	X	4.3.2.2, 4.4.4
3.6.1	Cover handle assembly				X		X	X	4.3.2.2, 4.4.5.1
3.6.2	Latch, cover and hinge assembly				X		X	X	4.3.2.2, 4.4.5.2
3.6.3	Cover hinge assembly				X		X	X	4.3.2.2, 4.4.5.3
3.6.4	Hasp assembly				X		X	X	4.3.2.2, 4.4.5.4
3.7.1	Gasket compression for M19A1				X	X	X	X	4.3.2.2, 4.4.6.1, 4.4.6.3, 4.4.6.4
3.7.2	Gasket compression for M2A1				X	X	X	X	4.3.2.2, 4.4.6.2, 4.4.6.3, 4.4.6.4
3.7.3	Gasket compression for M2A2				X	X	X	X	4.3.2.2, 4.4.6.2, 4.4.6.3, 4.4.6.4
3.8	Workmanship				X	X	X	X	4.5
Notes: 1/ For boxes coated in accordance with 12998172, an additional (First Article only) corrosion resistance test will be performed on 4 X 6 inch panels as part of verification class A in accordance with 4.4.2.2.									

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)

b. Conformance inspection (see 4.4)

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4.2 First article. When specified, a sample of the M19A1,M2A1, or M2A2 boxes shall be subject first article verification in accordance with Table II.

4.2.1 First article quantity. The first article sample shall consist of thirty (30) painted boxes and two (2) complete sets of unassembled component parts

4.2.2 First article rejection. If any item of the sample fails to comply with the first article requirements, the first article shall be rejected.

TABLE II. First article inspection

Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method
Assembly (Dwg. 7553296,7553315, or 13032542, as applicable) Examination for defects	15	3.2	4.3.2.1
Protective Coating 2/	3	3.3	4.4.1, 4.4.2, 4.4.2.1
Functioning	15	3.4	4.4.3
Air Tightness of box assembly	30	3.5	4.4.4
Gasket Compression M19A1 (Dwg. 7553315) M2A1 (Dwg. 7553296) M2A2 (Dwg. 13032542)	 10 10 10	 3.7.1 3.7.2 3.7.4	 4.4.6, 4.4.6.1, 4.4.6.3, 4.4.6.4 4.4.6, 4.4.6.2, 4.4.6.3, 4.4.6.4 4.4.6, 4.4.6.2, 4.4.6.3, 4.4.6.4
Weld Security 1/ Cover handle assembly Latch, cover and hinge assembly (Applicable to M19A1 box) Cover hinge assembly (Applicable to M2A1 and M2A2 boxes) Hasp assembly	 5 5 5 5	 3.6.1 3.6.2 3.6.3 3.6.4	4.4.5.1 4.4.5.2 4.4.5.3 4.4.5.4

NOTES:
1/ The weld security inspections shall be performed after the other inspections have been performed on the samples.
2/ For boxes coated in accordance with 12998172, an additional corrosion resistance test will be performed on three (3) 4 X 6 inch panels in accordance with 4.4.2.2.

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4.3 Conformance inspection.

4.3.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection should represent only the inspection lot from which drawn and should not be construed to represent any prior or subsequent quantities presented for inspection. The inspection lot should have been produced by one box assembly manufacturer. In addition, the inspection lot should consist of box assemblies in one unchanged process using the same drawings, revisions, materials and methods. All material submitted for inspection in accordance with this inspection should comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements. Lot numbering, as required, should be in accordance with MIL-STD-1168. Lot formation shall be in accordance with the lot formation requirements of MIL-STD-1916, paragraph "Formation and identification of lots and batches".

4.3.2 Sampling requirements. Inspection sampling requirements for Critical, Major and Minor characteristics are defined in MIL-STD-1916. Unless specified otherwise, Verification Level IV shall be used for all characteristics defined as Majors and Verification Level II for all Minor characteristics; Critical characteristics shall be addressed in accordance with MIL-STD-1916.

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4.3.2.1	Box, Ammunition, M2A1 Assembly Box, Ammunition, M19A1 Assembly Box, Ammunition, M2A2 Assembly			DRAWING NUMBER 7553296/7553315 13032542 NEXT HIGHER ASSEMBLY
Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method
Critical	None Defined.			
Major				
101	Inside length, minimum (min.)	VL IV	3.2	Gage
102	Inside width, min.	VL IV	3.2	Gage
103	Inside height, min.	VL IV	3.2	Gage
104	Incomplete manufacture	VL IV	3.2	Visual 1/
105	Incorrect assembly	VL IV	3.2	Visual
106	Incomplete or inadequate paint coverage	VL IV	3.2	Visual
107	Burrs, rough spots, sharp projections, cracks, splits, buckles	VL IV	3.8.1	Visual
Minor				
201	Outside length, maximum (max.)	VL II	3.2	AIE
202	Outside width, max.	VL II	3.2	AIE
203	Outside height, max.	VL II	3.2	AIE
204	Concavity or convexity of cover assembly	VL II	3.2	AIE
205	Concavity or convexity of body assembly	VL II	3.2	AIE
206	Chips, dirt, grease and other foreign matter	VL II	3.8.2	Visual
207	Surface imperfections, excessive paint not dry	VL II	3.2	Visual

NOTES:
1/ Refer to MIL-STD-406 for visual defects standards. The standard defects classified as incidental shall be considered permissible.

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4.3.2.2	Functioning/Airtightness/Gasket Compression/Coating/Weld Security			DRAWING NUMBER 7553296/7553315/ 13032542 NEXT HIGHER ASSEMBLY
Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method
Critical	None Defined.			
Major				
101	Airtightness	100% 1/	3.5	Visual/Manual
102	Gasket compression	VL IV	3.7	AIE
	Functioning			
103	Cover assembly	13 2/	3.4.1	Manual
104	Hasp and latch	13 2/	3.4.2	Manual
105	Elevated temperature	32 2/	3.4.3	Manual
106	Paint thickness and adhesion	3 3/	3.3.1	4.4.1
107	Corrosion resistance	3 3/	3.3.2	4.4.2, 4.4.2.1
	Weld security			
108	Cover handle assembly	13 2/	3.6.1	AIE
109	Latch, cover, and hinge assembly (M19A1)	13 2/	3.6.2	AIE
110	Cover hinge assembly (M2A1 and M2A2)	13 2/	3.6.3	AIE
111	Hasp Assembly	13 2/	3.6.4	AIE
Minor	None defined			

NOTES:
1/ Any box assembly which fails to meet the specific requirements shall be removed from the lot. These defective box assemblies may be reworked and resubmitted for retesting.
2 Failure of one or more sample unit to comply with the specified requirements shall be cause for rejection of the lot.
3/ Failure of any box to confirm to the requirement shall be cause to require the manufacturer to correct his painting process to comply with the requirement.

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4.3.2.3	Box, Shipping prior to sealing			DRAWING NUMBER 7692103/ 7553352 NEXT HIGHER ASSEMBLY
Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method
Critical	None defined			
Major	None defined			
Minor				
201	Contents incorrect	VL II	3.2	Visual
202	Separator missing	VL II	3.2	Visual

4.3.2.4	Box, shipping after sealing			DRAWING NUMBER 7692103/ 7553352 NEXT HIGHER ASSEMBLY
Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method
Critical	None defined			
Major				
101	Box damaged	VL IV	3.1	Visual
102	Contents of box exposed	VL IV	3.1	Visual
Minor				
201	Contents loose	VL II	3.1	Manual/Visual
202	Box inadequately or improperly sealed	VL II	3.1	Visual
203	Marking missing, incorrect or illegible	VL II	3.1	Visual

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4.4 Test methods and procedures.

4.4.1 Film thickness and paint adhesion. The method of test shall comply with applicable requirements of TT-C-490. The test shall be performed on three randomly selected surfaces of each unit of the sample.

4.4.2 Corrosion resistance . The method of test shall be as specified in ASTM B117 Method of Test for Salt Spray (FOG) Testing.

4.4.2.1 Assembled Box Corrosion Resistance. Using a sharp instrument, (machinist scriber or equivalent) one vertical line approximately three (3) inches long shall be scored across the flat portion of each side, the bottom, the hinge end, and the cover. The sample shall be positioned so that the hinged edge of the assembled box is elevated approximately 15 degrees from the horizontal and rests upon a wooden support while the cover face is parallel to the principal direction of horizontal flow of fog through the chamber. After salt spray testing, the scribe shall be evaluated according to ASTM D1654.

4.4.2.2 Test Panel Corrosion Resistance. Three (3) 4 X 6 inch test panels shall be prepared in accordance with 3.3.1 using the same steel and the same coating process used to manufacture the assembled box. Using a sharp instrument, (machinist scriber or equivalent) one vertical line approximately three (3) inches long shall be scored across one face of the panel. The panels shall be positioned on an angle such that scribed surface is coming into contact with the salt solution droplets. After salt spray testing, the scribe shall be evaluated according to ASTM D1654.

4.4.3 Functioning. The box shall be inspected to assure compliance with requirements of 3.4 through the following procedures:

a. Unlock and lock the cover assembly by hand leverage. Note the locking and unlocking action on the M19A1,M2A1, and M2A2 box assembly for effectiveness and ease of operation. Then only the M19A1 box assembly shall be unlocked, latch disengaged, and held in an inverted position. The formed lip inside of each M19A1 box assembly cover skirt shall catch beneath the edge of the body fold and restrain the cover from opening fully when partially opened box is inverted.

b. Remove and replace the cover assembly. Note any sticking of gaskets to the top edge of the body assemblies, or any misfit or bind within the body assembly and inspect hinge pins for security in the body half of the hinge.

c. Remove all cover assemblies. Inspect the gaskets for fit security, cuts, and splits.

d. Environmentally condition the box assemblies as specified in 3.4.3 after conditioning, re-inspect the box assemblies in accordance with a, b, and c above.

4.4.4 Airtightness of box assembly. The box shall be conditioned to a temperature of not less than 50°F and ambient pressure prior to testing. The boxes shall be tested in accordance with 4.4.4.1. A wetting agent may be used to minimize air bubbles clinging to the exterior surface (see 6.6).

4.4.4.1 Vacuum method. The assembled box shall be tested for leakage by immersing the closed box, in a vacuum vessel, to a depth of at least one inch below the surface of the water. The air pressure in the vessel shall be lowered to 3 psi below ambient pressure. Observation for leakage of air from the interior of the box assembly shall be made during the period of lowering pressure and for a minimum of 16 seconds while the 3 psi pressure differential is maintained.

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4.4.4.2 Hot water method for airtightness. The hot water procedure for each box should be submitted to the Government for approval. The test procedure should include water temperature, immersion time, the data to support time and periodic verification procedure. The assembled box shall be tested for leakage by immersing the closed box to a depth of one inch below the surface of the water. The temperature of the water and the length of time of immersion should be that which will assure an increase in pressure to a minimum of 3 psi above ambient pressure. Observation for air leakage from the interior of the box assembly should be made during the period of pressure build up and for a minimum of 16 seconds after the 3 psi pressure differential has been reached.

4.4.4.3 Dry vacuum method for airtightness. The dry vacuum test procedure should be submitted to the Government for approval. The test procedure should include periodic verification procedure and test device specification. The assembled box should be tested for leakage in a dry vacuum chamber by using an Army Peculiar Equipment (APE) 1958 test device or equal. Air leakage should not exceed .4 cubic centimeters for a minimum of 5 seconds.

4.4.5 Weld security.

4.4.5.1 Cover handle assembly. The assembled box shall be clamped with the sample body assembly resting on its bottom. Attach the test fixture to the cover handle. With one fixture securely held, apply the specified tensional force through the other test fixture, vertical to the box bottom against the ferrule of each handle. The force shall be applied at a rate of .125 to .25 inch per minute and held for one minute. After removal of the force, examine the handle and hasp for distortion and weld failure.

4.4.5.2 Latch, cover, and hinge assembly (M19A1). The assembly shall be tested in a suitable fixture with the box resting on its bottom and with the cover raised at a right angle. The box shall be clamped securely and the body assembly shall be supported by means of a snug fitting wood filler box in order to prevent its distortion or collapse. Secure the other test fixture to the latch assembly and slowly apply a tensional force perpendicular to the body assembly. The force shall be applied at a rate of .125 to .25 inch per minute and held for one minute. After removal of the force, examine the latch, latch link, latch link retainer, cover, cover hinge and body hinge for distortion and weld failure.

4.4.5.3 Cover hinge assembly (M2A1 and M2A2). The assembly shall be tested in a suitable fixture with the box resting on its bottom with the cover raised at a right angle. The box shall be clamped securely and the body assembly shall be supported by means of a snug fitting wood filler box in order to prevent its distortion or collapse. Secure the other test fixture to the cover and slowly apply a tensional force perpendicular to the body assembly. The force shall be applied at a rate of .125 to .25 inch per minute and held for one minute. After removal of the force, examine the cover, cover hinge and the body hinge for distortion and weld failure.

4.4.5.4 Hasp assembly. The sample body assembly resting on their bottoms shall be clamped in a suitable device. Using a test fixture clamped securely on the hasp assembly, the specified force shall be slowly applied perpendicular to the bottom and against the underside of the offset of the hasp. The force shall be applied at a rate of .125 to .25 inch per minute and held for one (1) minute. After removal of the force, examine the hasp for distortion and weld failure.

4.4.6 Gasket compression. (see dwg. 7553315/ 7553296/13032542 as applicable) .

4.4.6.1 Compressed gasket and cover measurements (Applicable to M19A1). The box shall be positioned in a suitable fixture so that it cannot be moved (see 6.3). For the M19A1, four points, two on each side, two inches from the front and the rear, shall be located (and numbered 1, 2, 3, and 4) on both the body and cover. With the body fully closed, a reading at each point, with a standard height gage, shall

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be recorded, then the cover shall be removed and the height to the hemmed edge of the body at the same point shall be recorded. Subtraction of the second reading from the first reading gives the value of the compressed gasket and cover.

4.4.6.2 Compressed gasket and cover measurements (Applicable to M2A1 and M2A2). For the M2A1 and M2A2, four points, two on each side, two inches from the front and the rear of the box body shall be located (and numbered 1, 2, 3 and 4) on both the body and the cover. With the body fully closed, a reading at each point (4 points for the M2A1 and M2A2), shall be recorded with a standard height gage, then the cover shall be removed and the height to the hemmed edge of the body at the same point shall be recorded. Subtraction of the second reading from the first reading gives the value of the compressed gasket and cover.

4.4.6.3 Uncompressed gasket and cover measurements. The cover and uncompressed gasket shall be measured at each of the four locations (points 1, 2, 3 and 4) for the M19A1, M2A1, and M2A2 with a precision measurement device and readings recorded. Care must be exercised so that the surface of the measuring device does not rest in the groove of the compression set. These readings are the uncompressed gasket and cover values.

4.4.6.4 Average gasket compression value. The compressed gasket and cover results (see 4.4.6.1 and 4.4.6.2) shall be subtracted from the uncompressed gasket and cover readings (see 4.4.6.3) at corresponding location points (1, 2, 3, and 4 only for both M19A1, M2A1, and M2A2). The sum of the resulting values for the four location points divided by four shall be considered the average gasket compression value.

4.5 Workmanship. Visually verify that all parts and assemblies meet the requirements of paragraph 3.8.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DOD or in-house personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Boxes, M19A1, M2A1, and M2A2 are intended for diversified use in the packaging of small arms ammunition. The M2A1 and M2A2 boxes is also suitable for the packaging of Artillery Fuzes, Rocket Fuzes, Mortar Fuzes, 60MM and 40MM cartridges and other ammunition.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification

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b. Quantity required and delivery schedules.

c. Serialization requirements, if applicable.

d. Quality conformance inspection, if other than specified in Section 4 of this specification.

e. First article sample requirements, if other than specified in Section 4 of the specification.

f. Packaging requirements, if other than specified in section 5 of this specification. Referenced drawings 7553352 for M2A1 and M2A2 boxes or 7692103 for M19A1 box.

g. Renovation, inspection, and packaging and palletization procedures, if renovation per Appendix A is applicable a contract data requirement DD Form 1423 for paragraph A4.2.

6.3 Measurement for gasket compression. Three point contact of the box resting on blocks may be used to overcome any rocking which might be encountered with the box resting on the surface plate. However, since the measurement technique relies upon relative distances, care should be taken to assure that there is no displacement on the referenced surfaces between the readings.

6.4 Inspection Equipment. The inspection equipment required to perform the examination and tests prescribed herein is described in the Inspection Method Column in the tables starting with table II, Inspection equipment used for acceptance of product should be capable of repeatable measurements, by various experienced inspection/test personnel, to an accuracy of 10% or better of the total tolerance of the characteristic being inspected. Any automated acceptance inspection equipment (AAIE) prove-out should be performed in accordance with MIL-A-70625.

6.5 Definitions.

6.5.1 Inspection Lot. The term “inspection lot” is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements.

6.6 Alternate test method for airtightness of box assembly. The alternate test method for airtightness of box assembly should be conducted as described in 4.4.4.2 or 4.4.4.3 with government approval.

6.7 Subject term (key word) listing.
M19A1, Small Arms Box
M2A1, Small Arms Box
M2A2, Small Arms Box

6.8 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue..

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APPENDIX A

RENOVATION OF M2A1/M2A2 AMMUNITION BOX ASSEMBLY PROVISIONS FOR RENOVATION OF M2A1/M2A2 AMMUNITION BOX ASSEMBLY

A1. SCOPE

A1.1 Scope. This appendix provides the Quality Assurance Provisions for renovation of Box, Ammunition, M2A1 Assembly and Box, Ammunition, M2A2 Assembly.

A2. REQUIREMENTS

A2.1 Seams. Defective seams should not be repaired. Box assemblies with this defect should be rejected and properly disposed of.

A2.2 Gaskets. The gasket on the cover assembly should be replaced.

A2.3 Marking. The renovated body assembly should be marked in accordance with Dwg. 7553297 or Dwg. 13032548 as applicable.

A2.4 Dents. Dents which do not preclude good stenciling and/or do not affect proper functioning of the box assembly or hinder proper packing of ammunition should be acceptable. The box assembly should not contain any dents greater than ¼ inch in depth. Dents should be removed as required to comply with the provisions described herein.

A2.5 Cleaning. There should be no rust present and the interior and exterior of the box assembly should be free of chips, dirt, grease, rust and foreign matter. All rust should be removed and the exterior and interior of the box assembly cleaned.

A2.6 Refinishing. Bare metal should be primed and finish coated as specified on the applicable drawings and 3.3.1. Bare metal should be primed and the complete exterior of the box assembly repainted. The interior of the body assembly and the inside of the cover assembly may be spot primed and repainted as required.

A2.7. Protective coating.

A2.7.1 Paint. See 3.3.1.

A2.7.2 Corrosion resistance. See 3.3.2.

A2.8 . Functioning.

A2.8.1 Cover assembly. See 3.4.1.

A2.8.2 Hasp and latch. See 3.4.2.

A2.8.3 Elevated temperature storage. See 3.4.3.

A2.9. Airtightness. See 3.5.

A2.10 Workmanship. See 3.8

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APPENDIX A

A3. VERIFICATION.

A3.1 Conformance inspection.

A3.1.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of renovated box assemblies from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection should represent only the inspection lot from which drawn and should not be construed to represent any prior or subsequent quantities presented for inspection. The inspection lot should have been produced by one box assembly manufacturer. In addition, the inspection lot should consist of box assemblies renovated in one unchanged process using the same drawings, revisions, materials and methods. All material submitted for inspection in accordance with this inspection should comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements. Lot numbering, as required, should be in accordance with MIL-STD-1168. Lot formation shall be in accordance with MIL-STD-1916.

A3.1.2 Examination. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans and procedures for major and minor defects should be in accordance with MIL-STD-1916, Verification Level IV for major defects and Verification Level II for minor defects, except that inspection for critical defects should be 100 percent. Continuous sampling plans in accordance with MIL-STD-1916 may be used if approved by the procuring activity.

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APPENDIX A

A3.1.2.1	Box, Ammunition, M2A1 Assembly Box, Ammunition,M2A2 Assembly			DRAWING NUMBER 7553296 / 13032542 NEXT HIGHER ASSEMBLY
Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method
Critical	None Defined.			
Major				
101	Airtightness	100%	3.5	4.4.4
102	Seam weld cracked	100%	3.5	Visual
103	Paint thickness	VL IV	A2.7.1	4.4..1
104	Paint adhesion	VL IV	A2.7.1	4.4.1
105	Corrosion Resistance	VL IV	A2.7.2	4.4.2
106	Functioning – cover assembly	VL IV	A2.8.1	4.4.3
107	Functioning – hasp and latch	VL IV	A2.8.2	4.4.3
108	Functioning – elevated temperature storage	VL IV	A2.8.3	4.4.3
109	Gasket compression	VL IV	A3.7	4.4.6
110	Dents affecting stenciling, functioning or packing of ammunition or dents greater than ¼”	VL IV	A2.4	Visual/SMTE
111	Cleaning improper prior to priming and painting	VL IV	A2.5	Visual
112	Gasket not replaced properly	VL IV	A2.2	Visual
113	Bare metal not primed prior to painting	VL IV	A2.6	Visual
114	Paint improper	VL IV	3.3.1	Visual
115	One or more spot or projection welds broken on cover assembly or body assembly	VL IV	3.4	Visual
Minor				
201	Marking on body assembly missing, incomplete or illegible	VL II	3.4	Visual
202	Evidence of poor workmanship	VL II	3.8	Visual

A4. Notes.

A4.1 The renovated box assemblies should be packaged in accordance with dwg. 7553352 or procedures approved by the Government for packaging and palletization.

A4.2 Submission of renovation, inspection, and packaging and palletization of procedures. The procedures for the renovation, inspection, and packaging and palletization of the M2A1 and M2A2 Ammunition Box Assemblies should be submitted to the Government for approval.

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